IT PROFESSIONAL PORTFOLIO

Windows Infrastructure and Administration Projects

PROJECT 05

Domain Join and Computer Account Management

Windows Server 2025 | Windows 11 Pro | AD DS | DNS | Secure Channel

AUTHOR	Michael Salazar
ENVIRONMENT	Personal Windows Enterprise Lab
VERSION	v1.0.0
RELEASE DATE	August 14, 2026
STATUS	Complete

Executive Summary

Integrated a Windows Server 2025 member server and Windows 11 management workstation into the **corp.mslab.test** Active Directory domain. The project configured isolated-lab IPv4 and DNS settings, verified AD service discovery, joined both systems, authenticated domain identities, tested machine secure channels, and confirmed computer-object placement in purpose-built OUs.

Business Problem

A functioning forest cannot centrally authenticate or manage systems until endpoints can reliably discover domain services and establish trusted computer accounts. Incorrect addressing or external DNS prevents Kerberos, LDAP, Group Policy, and domain joins from operating predictably.

Project Objectives

- Protect both guests with pre-join Hyper-V checkpoints.
- Assign stable addresses on the isolated 172.16.10.0/24 network.
- Use CORP-DC1 as the exclusive AD DNS server.
- Validate the domain, host, and LDAP SRV records before joining.
- Join CORP-SRV01 and MGMT-CL01 to corp.mslab.test.
- Prove domain authentication and machine secure-channel health.
- Confirm enabled computer accounts and correct OU placement from the domain controller.

RESULT	Both machines joined successfully, authenticated domain users, returned PartOfDomain = True, and passed secure-channel validation.

Environment and Addressing Plan

System	Role	IPv4 / prefix	DNS
CORP-DC1	Domain controller and AD DNS	172.16.10.10/24	Local AD DNS
CORP-SRV01	Windows Server 2025 member server	172.16.10.20/24	172.16.10.10
MGMT-CL01	Windows 11 Pro management client	172.16.10.30/24	172.16.10.10

Trust and Placement Design

Computer	Target OU	Administrative purpose
CORP-SRV01	Corp / Servers	Server policy and administration scope
MGMT-CL01	Corp / Desktops / Corporate	Corporate workstation policy scope

Technical Dependency

Active Directory clients locate domain controllers through DNS SRV records. Successful resolution of **_ldap._tcp.dc._msdcs.corp.mslab.test** was treated as a hard prerequisite. No router or default gateway was required because all three systems communicate on the same private Hyper-V subnet.

CONTROL	AD-aware DNS was configured before any join attempt, reducing the chance of partial or misleading domain-join failures.

Rollback Protection

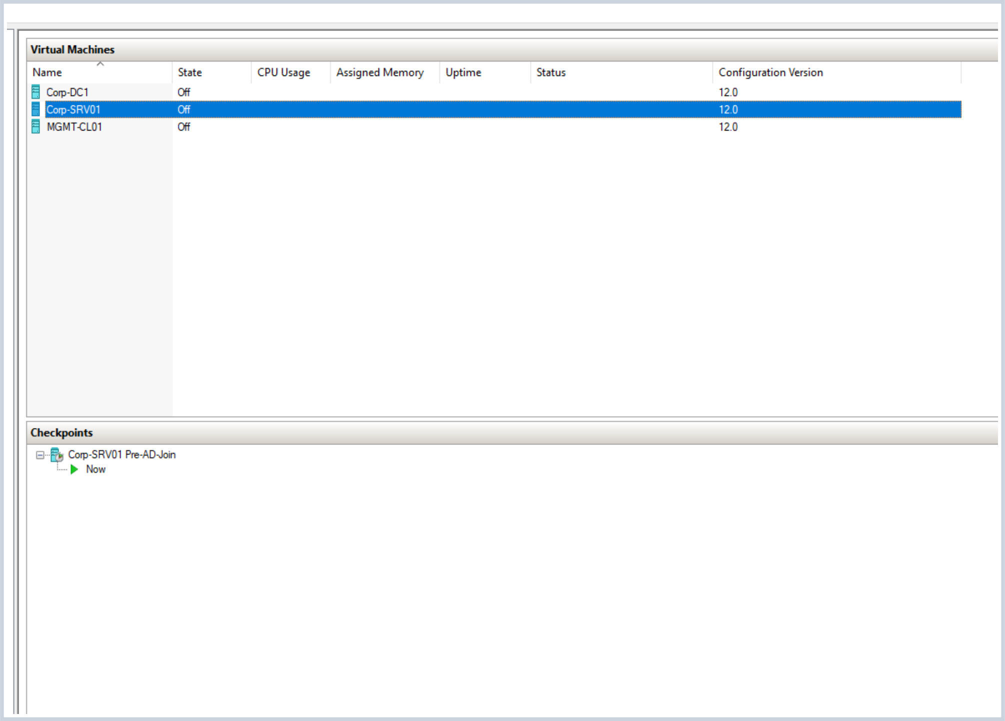

Figure 1 - CORP-SRV01 pre-join checkpoint

The server is protected with a clearly identified pre-Active Directory join rollback point.

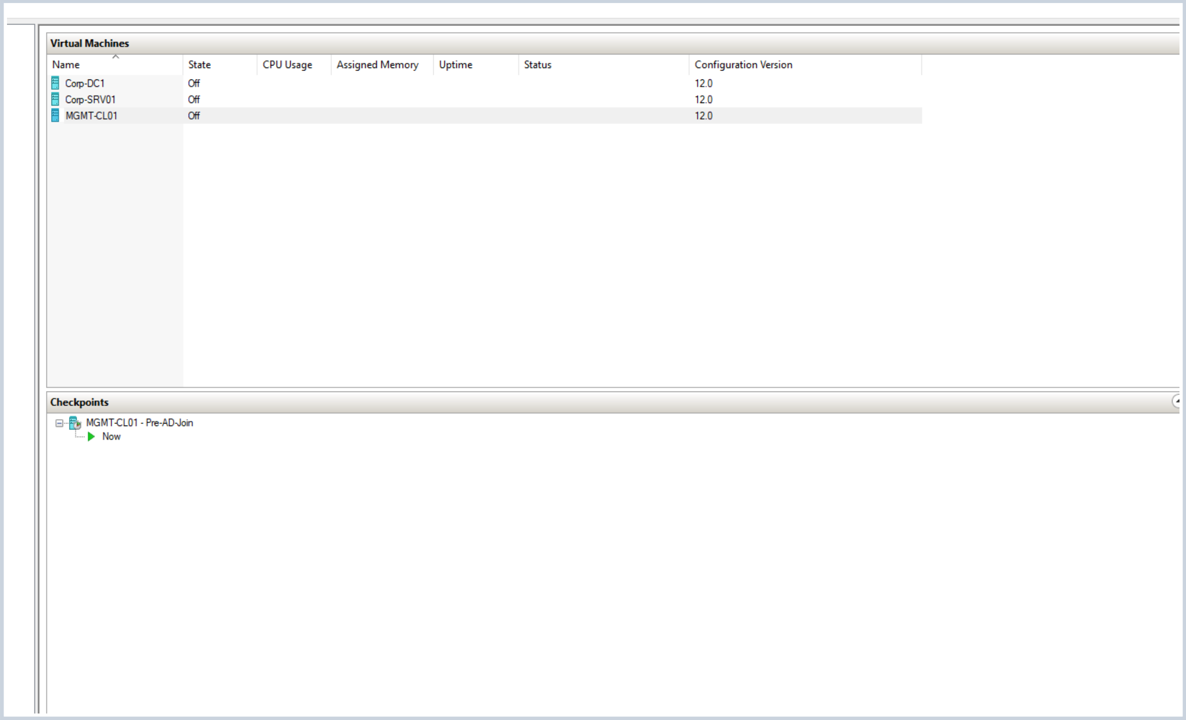

Figure 2 - MGMT-CL01 pre-join checkpoint

The Windows 11 client is shut down and protected before trust-state changes.

Network Troubleshooting and AD DNS Discovery

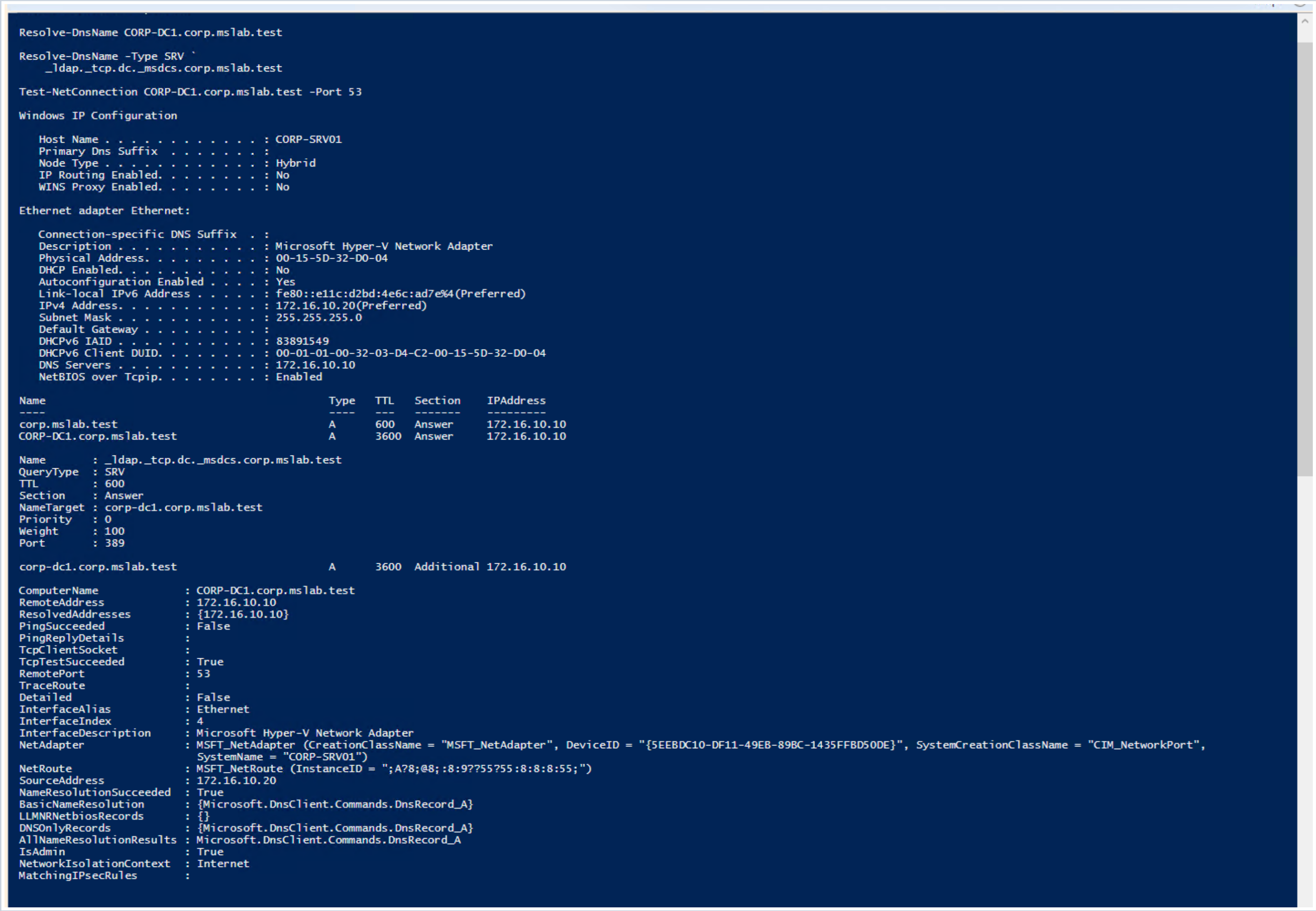

Figure 3 - CORP-SRV01 DNS and connectivity validation

PowerShell captures the corrected static configuration, connectivity to 172.16.10.10, DNS responses, LDAP SRV discovery, and TCP port 53 reachability.

The initial guest configuration used an APIPA address in the 169.254.0.0/16 range and IPv6 placeholder DNS servers, so domain queries timed out. CORP-SRV01 was corrected to 172.16.10.20/24 and MGMT-CL01 to 172.16.10.30/24; both were configured to query 172.16.10.10.

ROOT CAUSE	No DHCP service existed on the private Hyper-V switch. Static addressing and the domain controller DNS address restored service discovery.

PASS	corp.mslab.test and CORP-DC1 resolve to 172.16.10.10; the LDAP SRV record advertises CORP-DC1 on TCP 389.

CORP-SRV01 Domain Join

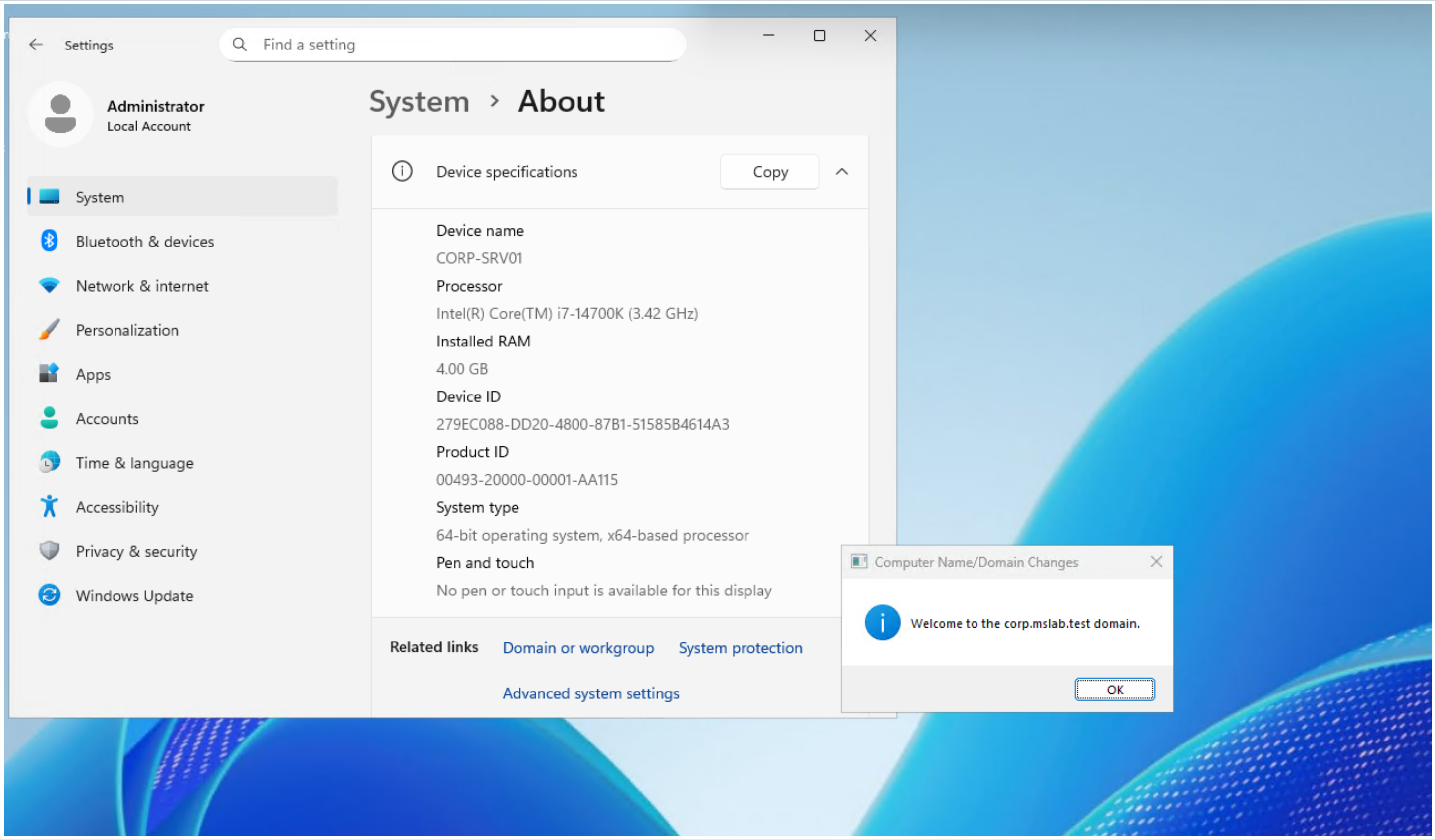

Figure 4 - Successful member-server join

Windows confirms that CORP-SRV01 joined the corp.mslab.test domain.

The server join creates a machine identity and a shared trust secret between CORP-SRV01 and Active Directory. The system was restarted to activate the new domain membership.

SECURITY	Server administration uses a domain administrative identity. The separate local account is named admin and is retained for break-glass/local recovery use.

Expected Enterprise Capabilities

The trusted member server can now receive Group Policy, use Kerberos and LDAP, resolve domain resources, and support later file-services, managed service-account, permission, and monitoring projects.

CORP-SRV01 Trust Validation

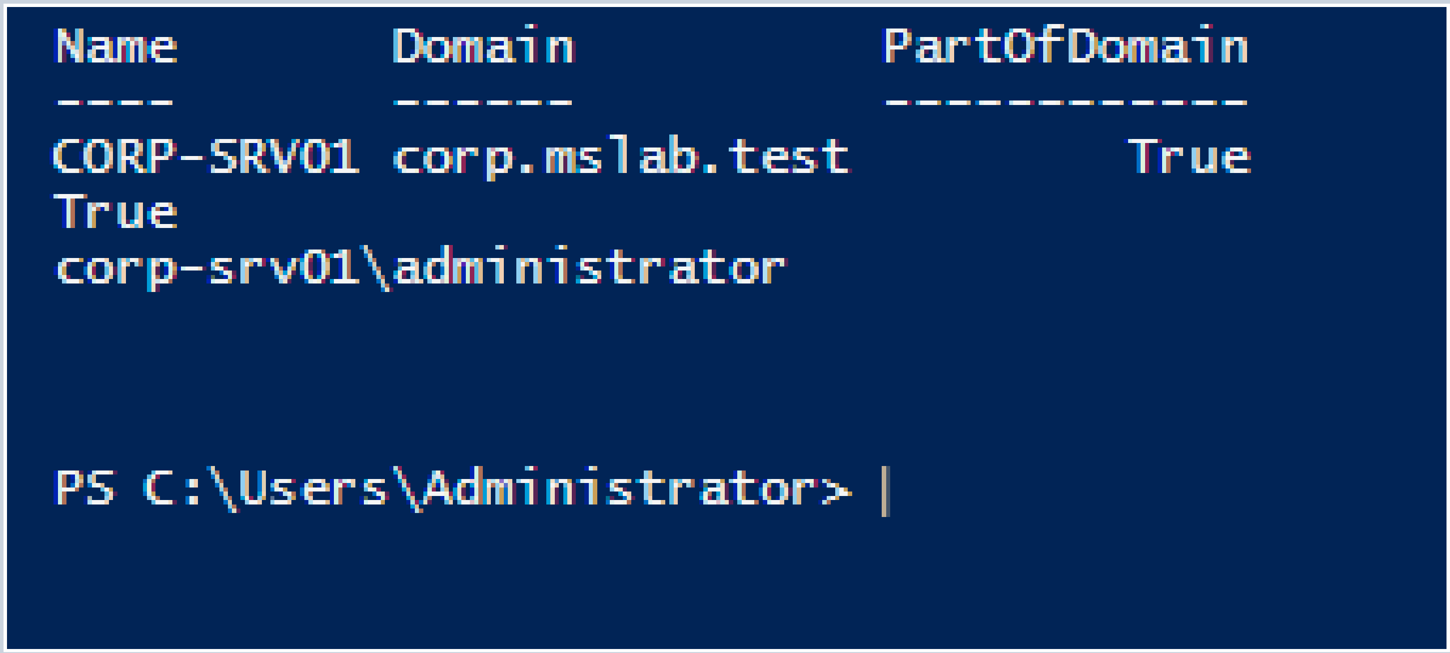

Figure 5 - Member-server validation

PowerShell confirms Domain = corp.mslab.test, PartOfDomain = True, and Test-ComputerSecureChannel = True.

Validation	Result
Computer name	CORP-SRV01
Domain	corp.mslab.test
Part of domain	True
Secure channel	True
Captured session	corp-srv01\administrator

The environment owner reports that the server was accessed with the domain administrator and that the deliberately created local recovery account is named **admin**. The captured **whoami** value is retained exactly as displayed. The authoritative machine-trust checks are the reported domain membership and successful secure-channel test.

PASS The member server has a healthy machine trust with CORP-DC1.

MGMT-CL01 Domain Join

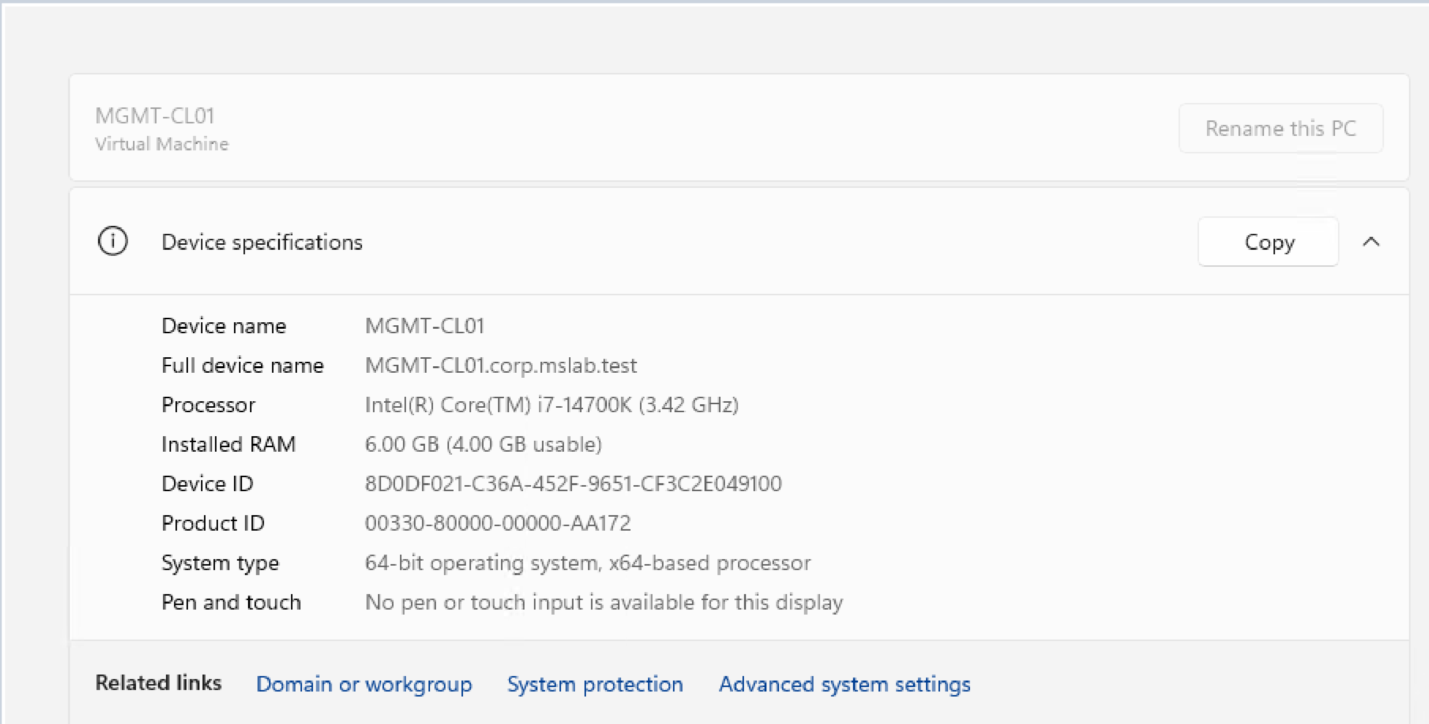

Figure 6 - Successful Windows 11 join

System settings confirm the management client joined corp.mslab.test and requires a restart.

MGMT-CL01 was joined using a delegated administrative credential after the same DNS prerequisites passed. The computer account was prestaged or placed beneath the corporate desktop hierarchy so workstation policy can be scoped independently of servers.

DESIGN	Separating servers and desktops by OU prepares the environment for role-specific Group Policy without mixing computer classes.

Authentication Test

After restart, the standard user Michael Salazar authenticated as CORP\msalazar. This proves more than object creation: the client can locate the domain, validate user credentials, and establish an interactive domain session.

MGMT-CL01 Authentication and Trust Validation

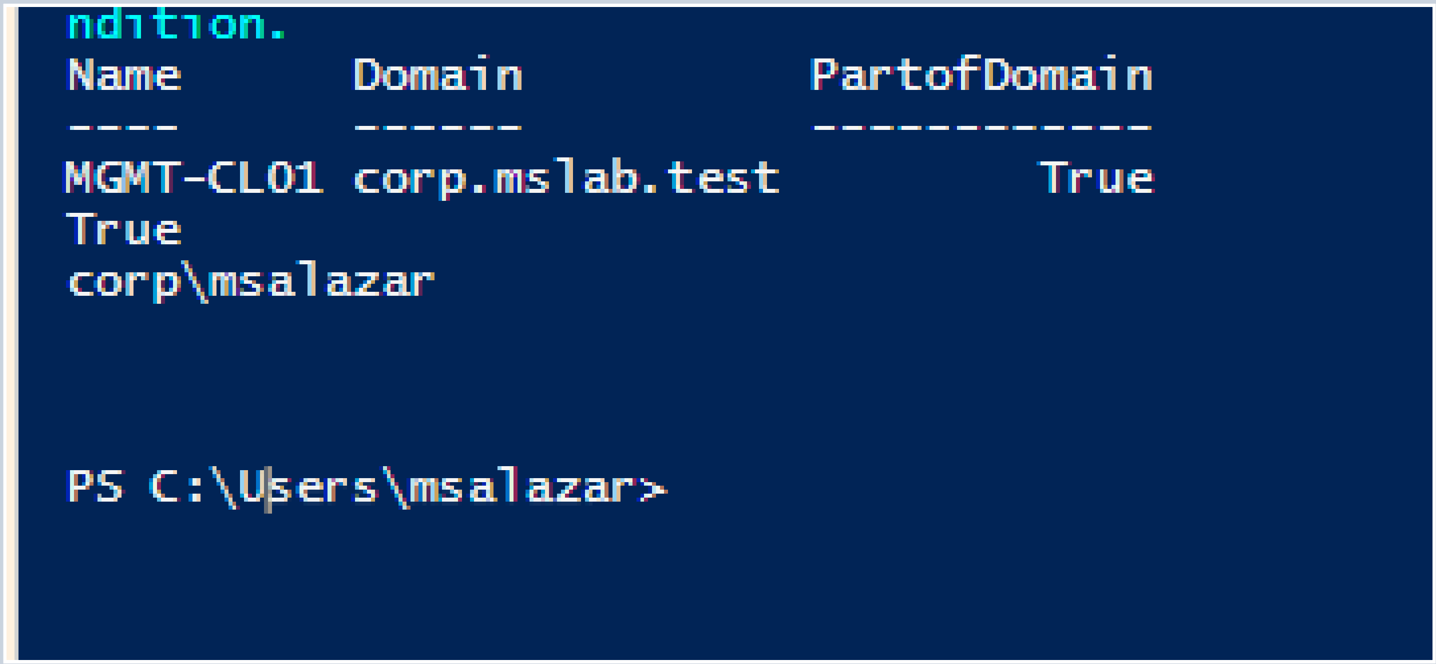

Figure 7 - Client-domain validation

PowerShell confirms the domain, PartOfDomain = True, secure channel = True, and interactive identity corp\msalazar.

Validation	Result
Computer name	MGMT-CL01
Domain	corp.mslab.test
Part of domain	True
Secure channel	True
Signed-in identity	corp\msalazar

Using the standard account for workstation access demonstrates separation between routine user activity and privileged server administration. The successful secure-channel test verifies the client computer account password and trust relationship remain synchronized with Active Directory.

PASS	Domain user authentication and workstation trust both validated successfully.

Central Computer-Object Validation

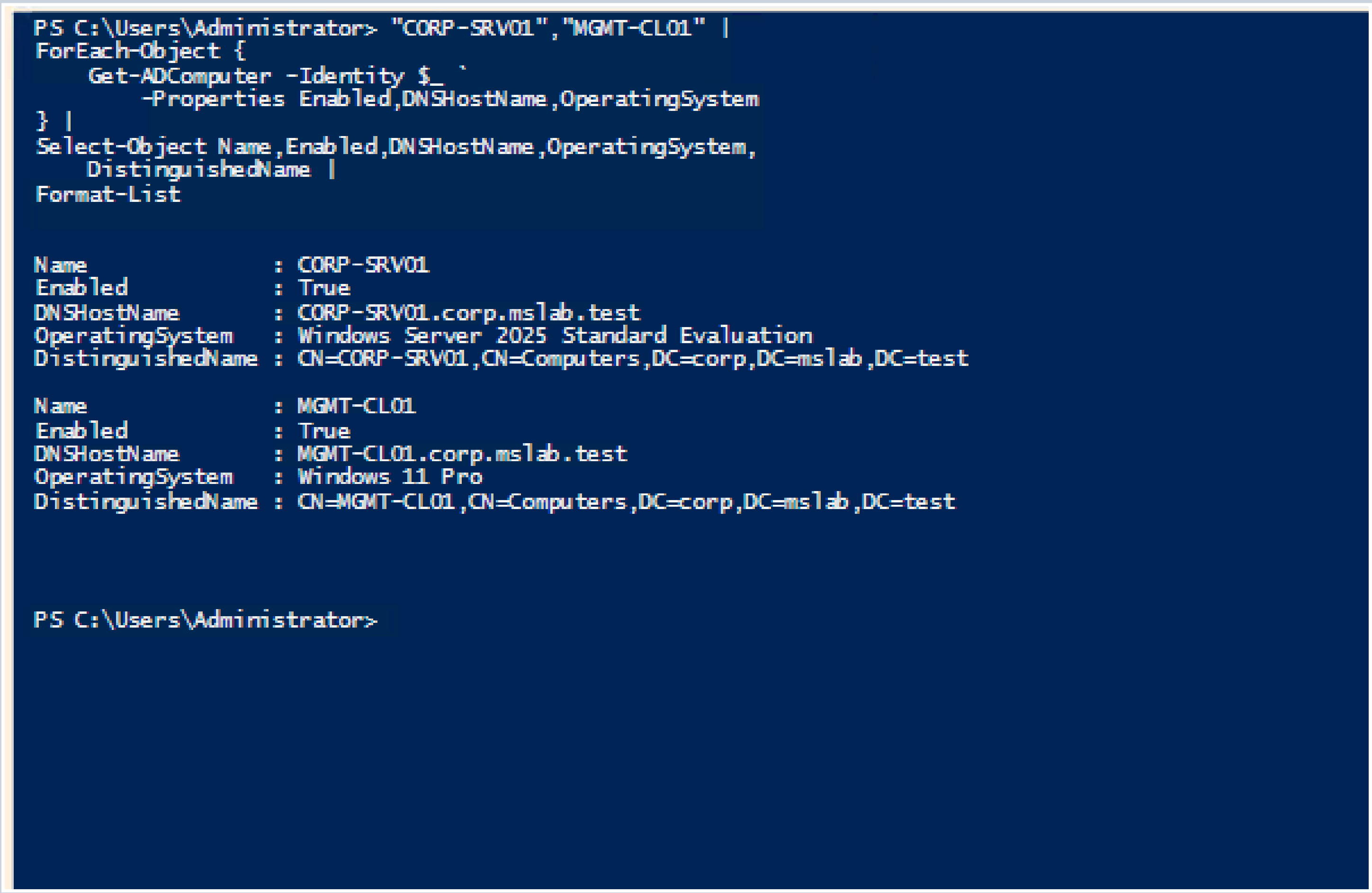

Figure 8 - Domain-controller validation

Get-ADComputer confirms both enabled objects, DNS hostnames, operating systems, and final distinguished names.

Computer	Enabled	Authoritative OU path
CORP-SRV01	True	OU=Servers,OU=Corp,DC=corp,DC=mslab,DC=test
MGMT-CL01	True	OU=Corporate,OU=Desktops,OU=Corp,DC=corp,DC=mslab,DC=test

This server-side check closes the validation loop: local systems report a healthy trust, while Active Directory independently reports enabled objects with correct DNS and operating-system metadata in the intended administrative scopes.

PASS	Both computer accounts are enabled and deliberately placed outside the default Computers container.

Validation Matrix

Control	Status	Evidence result
Rollback checkpoints	PASS	Pre-join checkpoints captured for both guests
IPv4 configuration	PASS	172.16.10.20/24 and 172.16.10.30/24
AD DNS	PASS	Both clients use 172.16.10.10
Domain discovery	PASS	A and LDAP SRV records resolve correctly
CORP-SRV01 join	PASS	PartOfDomain and secure channel are True
MGMT-CL01 join	PASS	PartOfDomain and secure channel are True
Domain authentication	PASS	Standard client session validated as corp\msalazar
Computer objects	PASS	Enabled, named, and placed in intended OUs

Resume Bullet

- Integrated Windows Server 2025 and Windows 11 systems into an Active Directory domain by remediating isolated-network addressing, configuring AD DNS, validating LDAP service discovery, establishing machine trusts, testing secure channels, and organizing computer objects into policy-ready OUs.

STAR Interview Story

STAR	Response
Situation	The member server and management client could not resolve the lab domain because the isolated Hyper-V network had no DHCP service.
Task	Establish reliable AD connectivity and join both systems without compromising directory placement or validation quality.
Action	Diagnosed APIPA addressing, assigned static IPv4 configurations, pointed both systems exclusively to AD DNS, validated host and LDAP SRV records, joined both machines, authenticated domain identities, tested secure channels, and audited the computer objects from CORP-DC1.
Result	Both machines now maintain healthy domain trusts and reside in purpose-built OUs ready for Group Policy, file services, remote administration, and security monitoring.

Version History

Version	Date	Change
v1.0.0	August 14, 2026	Initial completed release with network remediation, joins, trust validation, authentication, and OU evidence.